HPS Gaussian-Process Resonance Search
The Global-Significance Study

Full 2016 and 2021 10%: a v4.9.15 reader's report

Emrys Peets

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How to read these results. The study asks how often a complete background-only search would look this unusual somewhere on its mass grid. All probabilities condition on one archived background per dataset. They do not establish that background as physically correct, and they do not yet constitute a final particle-significance claim.

1 What has been extended

The 2015 study now has two companions: all of the 2016 dataset and the native 10% 2021 dataset. Each starts with ten complete simulated searches. Separate, larger simulations test the approximation before its scan-wide probabilities are interpreted. The original 2015 results and earlier calibration releases remain unchanged.

The new GP describes how the *likelihood-based search results at different masses move together*. The local signal-and-background likelihood is the same as before. This is a method for calculating a look-elsewhere correction, rather than a new simultaneous fit of the whole search.

Dataset	Mass [MeV]	Local p	GP global p	Direct exceedances
2015 full	51	4.09×10^{-4}	0.019	13/1000
2016 full	42	5.44×10^{-22}	$<1.50 \times 10^{-5}$	0/1000
2021 10%	92	2.78×10^{-4}	0.0252	26/1000

Table 1: Profiled statistic, principal minimum-local-p ordering. The 2015 row is reused from the frozen study. Each row is a separate search. A less-than sign gives a one-sided 95% Monte Carlo upper bound for a zero-count tail, conditional on the sampled model.

2016 full: at 42 MeV the observed root is 0.041, while the stress background predicts -9.215. The earlier two-background local envelope gives p=0.4044. The new extreme score measures tension with one background construction; it is not a particle discovery significance. The direct sample does not resolve this rare tail.

2021 10%: the GP estimate lies inside the direct-scan 95% interval at the observed threshold. This tests the approximation under that background, not the background itself.

2 How one simulated search becomes a global probability

Imagine scanning a long spectrum for a bump. Nearby mass tests share events and sidebands, so they are correlated. Counting every mass as an independent chance is inaccurate. The global calculation must reproduce those relationships.

First, simulate a whole experiment. Draw one complete spectrum from its stated no-particle background, then scan exactly those same counts at every mass. This preserves the shared fluctuations. The ten pilot scans are kept separate from the 1,000 independent validation scans; old mass-by-mass toy collections are not spliced together.

Second, measure the response of the search. Following the construction of Ananiev and Read [1], perturb each data bin of an unfluctuated background spectrum in turn. This estimates how fluctuations in that bin affect the scan. The resulting covariance describes correlations across the mass grid.

Third, draw inexpensive correlated scan curves. Generate 200,000 curves from that covariance and count how often any mass reaches the observed threshold. Independent full Poisson experiments test the approximation's means, spreads, correlations and maximum tails.

Dataset	Mass range [MeV]	Masses	Data bins	Asimov scans
2016 full	39–180	142	720	721
2021 10%	50–250	201	422	423

Table 2: Both mass grids have 1 MeV spacing. Each dataset also has ten pilot and 1,000 independent validation scans. One Asimov scan is the unperturbed mean; the remaining scans perturb one bin each.

The extension retains the nonzero background offset and the response width explicitly. If $r(m)$ is the signed likelihood-ratio root, write

$$a(m) = r(B; m), \qquad D_i(m) = r(B + \sqrt{B_i} e_i; m) - a(m), \qquad C = D^{\mathsf{T}} D,$$

and set $s_m = \sqrt{C_{mm}}$, $K_{mn} = C_{mn}/(s_m s_n)$. A sampled GP vector has covariance K; the corresponding raw-root curve is $a + sZ$. Thus the method does not silently assume that the original statistic is already centered at zero with unit width.

The principal rule uses $p_m = \overline{\Phi}[(r_m - a_m)/s_m]$ when $r_m > 0$, and $p_m = 1$ otherwise. It scans the smallest p_m. A separate raw-root maximum retains the original asymptotic ordering. These are different tests; their probabilities cannot be selected according to which gives the stronger-looking result.

3 Full 2016: the p-value scan

Profiled statistic. The most extreme point for the declared rule is 42 MeV. The local common-background probability is 5.44×10^{-22}, and the GP scan-wide estimate is $<1.50 \times 10^{-5}$. Direct scans exceed this threshold in 0/1000 cases, giving a one-sided 95% upper bound of 0.003.

Fixed statistic. The most extreme point for the declared rule is 43 MeV. The local common-background probability is 2.53×10^{-27}, and the GP scan-wide estimate is $<1.50 \times 10^{-5}$. Direct scans exceed this threshold in 0/1000 cases, giving a one-sided 95% upper bound of 0.003.

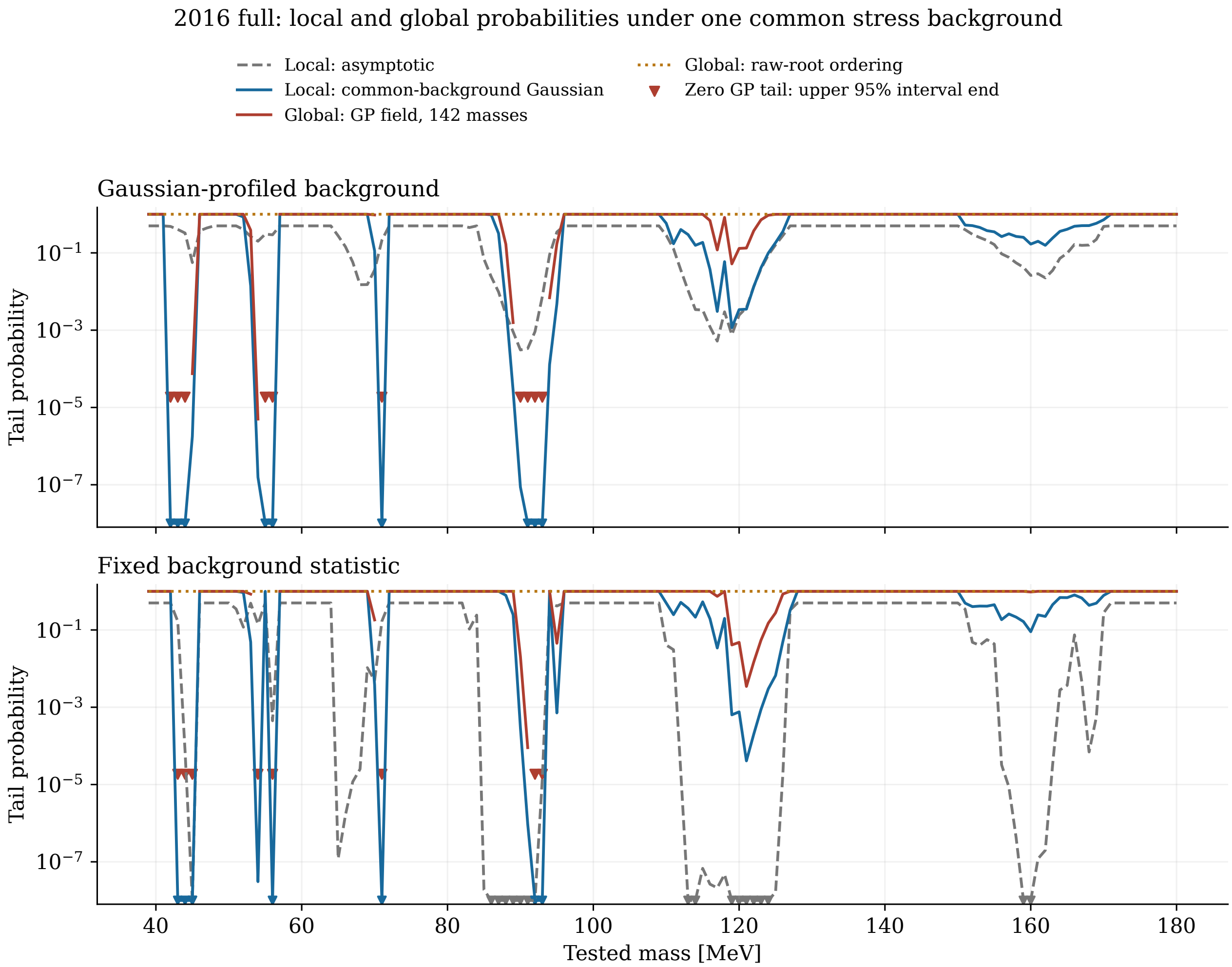

Figure 1: Full 2016 on its declared 142-point grid. The blue and red curves use the new common-background local-probability rule; the dotted curve uses the separate raw-root ordering. The gray curve retains the old asymptotic display, including its 0.5 convention for nonpositive roots; the calibrated rule assigns one there. Triangles at 10^{-8} mark smaller analytic probabilities. Zero simulated tails are shown at the upper endpoint of their 95% binomial interval, never as a measured zero probability.

Method	Mass	Observed r	Null offset a	Width s	$(r-a)/s$
Profiled	42	+0.041	-9.215	0.967	+9.568
Fixed	43	+0.965	-31.905	3.054	+10.765

Table 3: The peak of the principal rule, with its raw value and calibration ingredients. The standardized score is conditional on the common stress background; it is not an independently established Gaussian discovery significance.

4 Why the 2016 interpretation needs care

The selected stress background can itself cause the fitting procedure to infer large alternating positive and negative signals, even though no particle was injected. Centering the scan against that background changes which masses are most unusual. This is the same background-recovery issue exposed in the earlier calibration study.

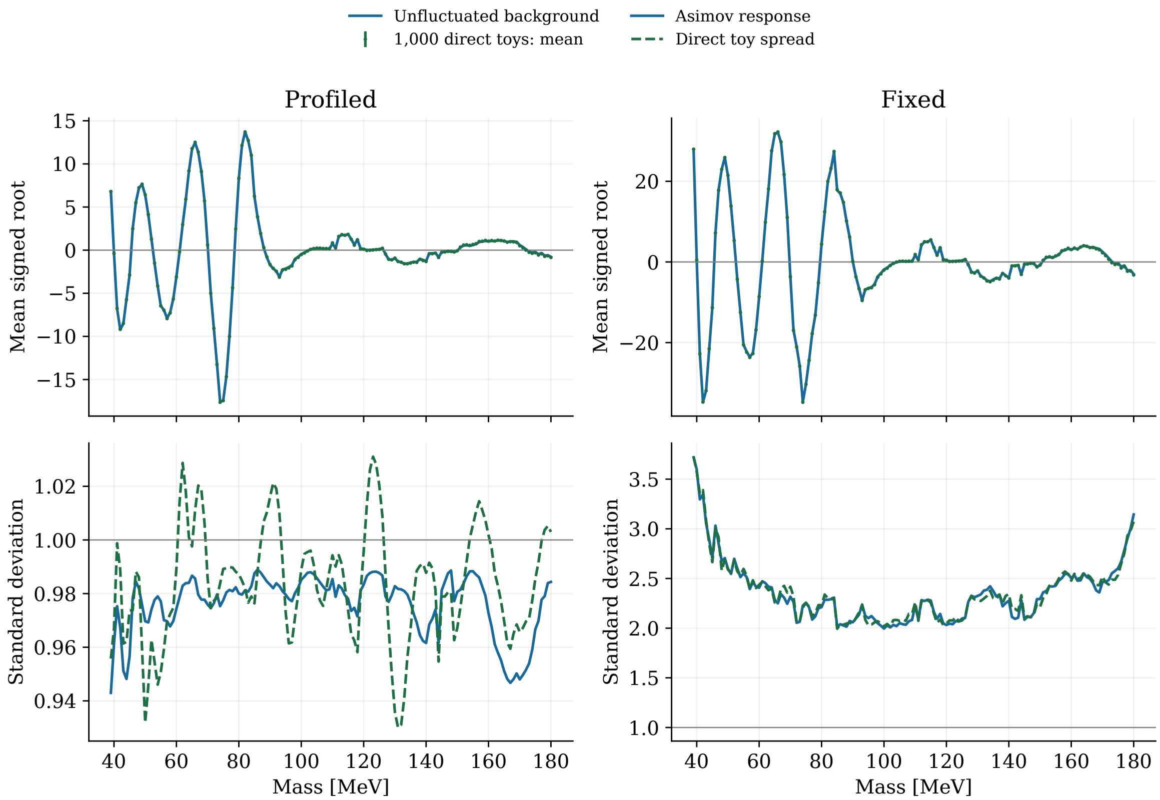

Figure 2: The unfluctuated-background prediction compared with 1,000 independent full-spectrum Poisson scans. The top panels show the offset in the raw signed root; the bottom panels show its spread. Agreement assesses the response approximation under the selected generating spectrum. It does not qualify that spectrum as the physical background.

Method	Standardized mean range	Spread range	Flagged masses
Profiled	-0.099 to +0.072	0.947 to 1.047	0
Fixed	-0.120 to +0.080	0.960 to 1.051	0

Table 4: Independent validation after deterministic centering and scaling. Normality flags use a Holm adjustment over 142 masses within each method. Non-rejection has finite statistical power and does not certify extreme tails.

The response uses the unfluctuated spectrum and its bin perturbations; the independent validation sample is not used to retune the offset, width or covariance.

The 2016 source was built from a pre-existing 10% development subset and joined to an archived broad-tail continuation over 75–85 MeV. Event-level independence from the full data was not established. The continuation retains a failed fit-status flag, with a waiver only for conditional stress testing. The inherited 2016 numerical exception is also unresolved by this extension. These qualifications remain attached to every 2016 number here.

5 2021 at 10% exposure: the p-value scan

Profiled statistic. The most extreme point for the declared rule is 92 MeV. The local common-background probability is 2.78×10^{-4}, and the GP scan-wide estimate is 0.0252. Direct scans exceed this threshold in 26/1000 cases (95% interval 0.0171–0.0379).

Fixed statistic. The most extreme point for the declared rule is 77 MeV. The local common-background probability is 1.76×10^{-6}, and the GP scan-wide estimate is 2.05×10^{-4}. Direct scans exceed this threshold in 1/1000 cases (95% interval 2.53×10^{-5}–0.0056). This small direct count gives only a weak rare-tail check; agreement with its broad interval is not precise validation of the GP estimate.

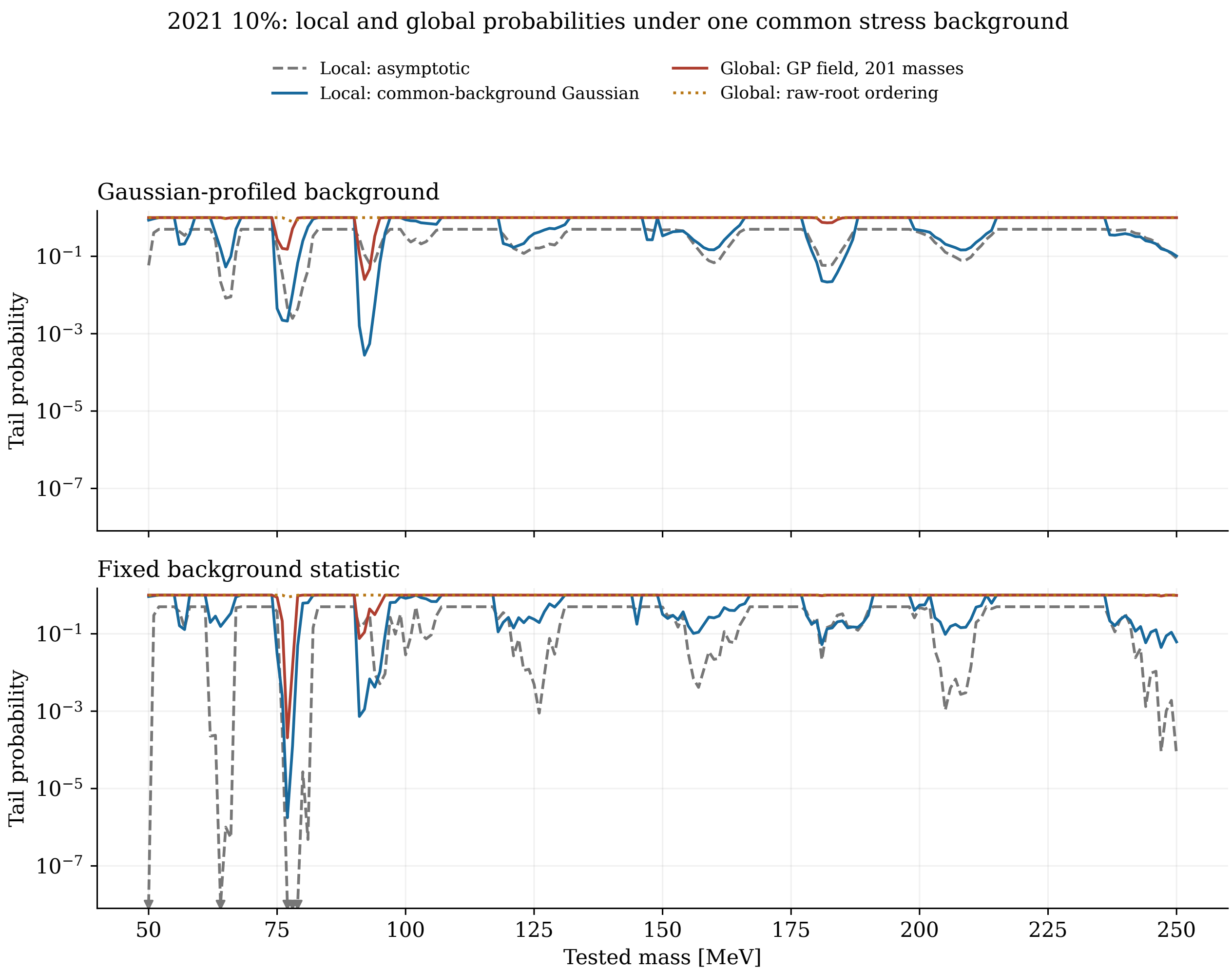

Figure 3: Native 2021 10% data on its declared 201-point grid. Definitions and markers match Figure 1. Both statistics use the same complete toy spectra. This study does not rescale the result to full 2021 exposure and does not include trials for choosing between years or analysis methods.

Method	Mass	Observed r	Null offset a	Width s	$(r-a)/s$
Profiled	92	+1.222	-2.177	0.984	+3.452
Fixed	77	+8.378	-1.809	2.197	+4.638

Table 5: The peak of the principal rule, with its raw value and calibration ingredients. The standardized score is conditional on the common stress background; it is not an independently established Gaussian discovery significance.

6 Checking the 2021 response approximation

The 2021 calculation uses the archived native-10% smooth background and the already selected 36–300 MeV GP support. Its search interval remains 50–250 MeV. The background is held fixed as a generating model, while toy fits retrain their log-count means and count-dependent errors. The observed kernel settings remain frozen.

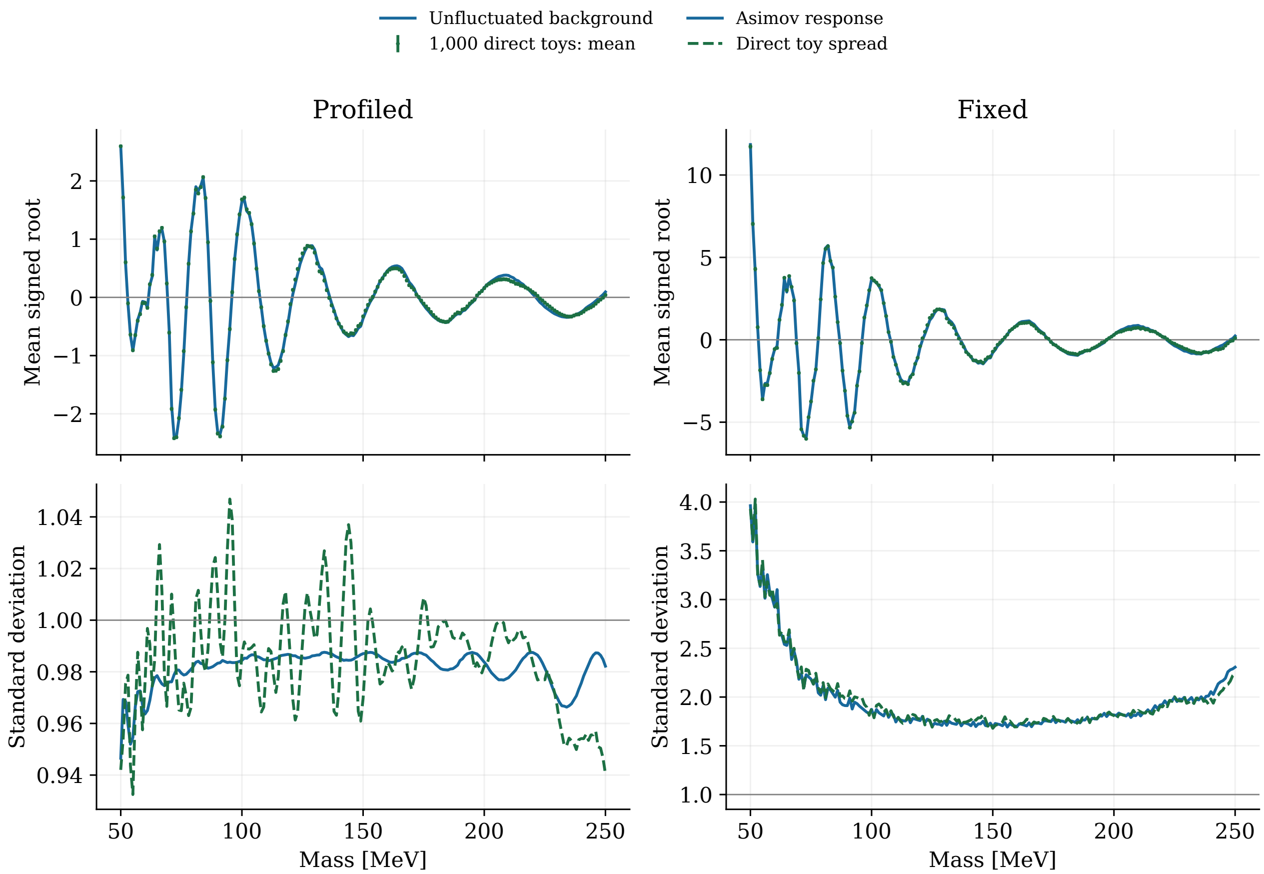

Figure 4: The same mean-and-width checks for 2021 10%. A narrower raw fixed-background fit does not guarantee a unit-width sampling distribution: fluctuations in the estimated background remain present across repeated experiments.

Method	Standardized mean range	Spread range	Flagged masses
Profiled	-0.079 to +0.087	0.957 to 1.064	0
Fixed	-0.093 to +0.083	0.943 to 1.057	0

Table 6: Independent validation after deterministic centering and scaling. Normality flags use a Holm adjustment over 201 masses within each method. Non-rejection has finite statistical power and does not certify extreme tails.

The response uses the unfluctuated spectrum and its bin perturbations; the independent validation sample is not used to retune the offset, width or covariance.

The earlier support study established practical recovery under its source-conditioned controls, with disclosed exceptions. It was not a proof of exact closure for every mass or every possible background. The present global study tests a different, fixed-kernel sampling procedure and retains that distinction.

7 How neighboring mass tests are related

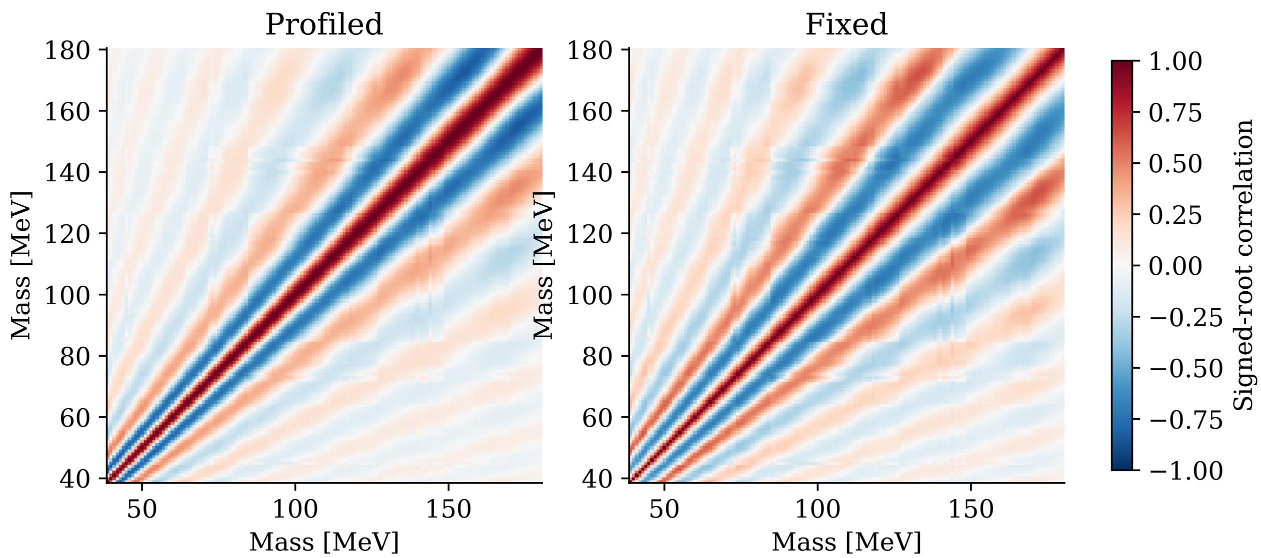

Figure 5: The 2016 response correlation matrices. Red entries mean that two tested masses tend to move in the same direction; blue entries mean opposite motion. Each diagonal entry is one by construction. This pattern replaces an assumption of independent mass tests.

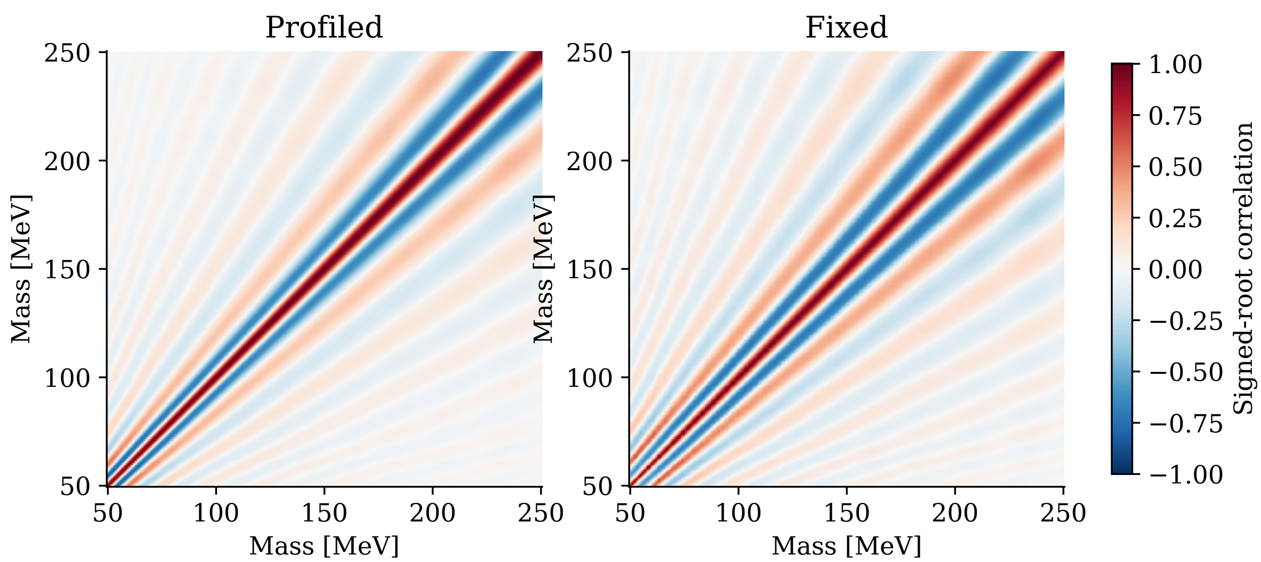

Figure 6: The corresponding 2021 matrices. The fixed and profiled statistics can have different correlation patterns despite using identical input spectra. These are correlations of search statistics, not correlations of physical signal production.

2016 full: the root-mean-square difference between the response correlation and the direct-scan correlation is 0.028 (profiled) and 0.031 (fixed).

2021 10%: the root-mean-square difference between the response correlation and the direct-scan correlation is 0.030 (profiled) and 0.028 (fixed).

8 Do the GP curves reproduce complete Poisson experiments?

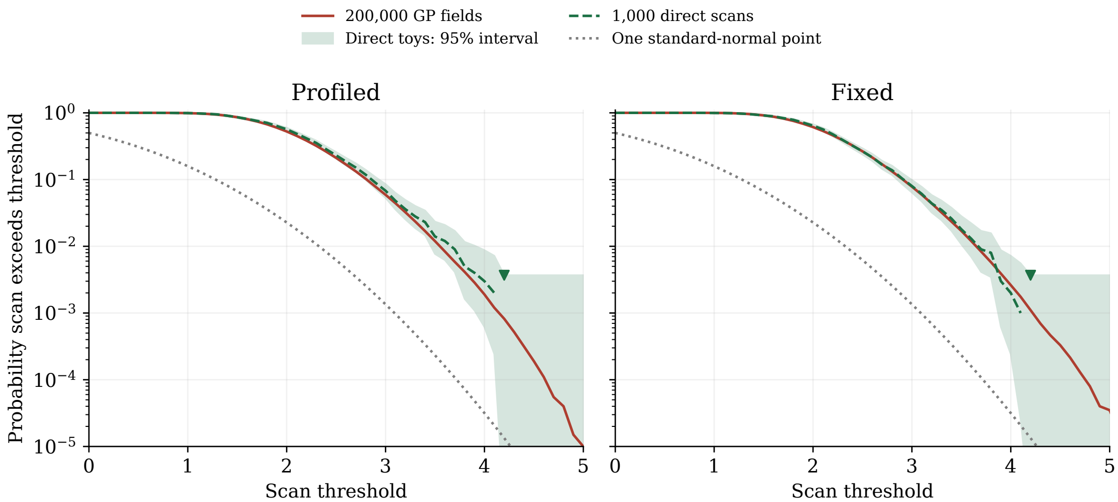

Figure 7: 2016 scan-maximum tails for the principal minimum-local-p ordering. Green bands are pointwise 95% binomial intervals from 1,000 direct scans; the red curve uses 200,000 GP draws. Triangles mark the first unresolved zero-count tail. The vertical range deliberately focuses on probabilities that the direct sample can begin to test.

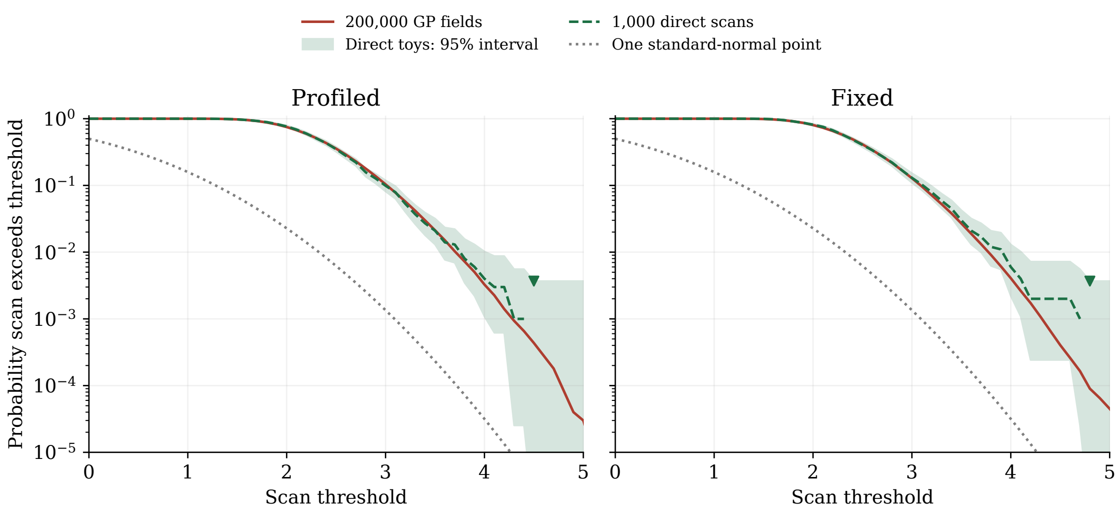

Figure 8: The equivalent validation for 2021. Agreement in the displayed range does not validate arbitrarily rare tails. The bands are pointwise sampling intervals, not a simultaneous coverage statement for the entire curve.

2016 full: the distribution-shape (KS) diagnostic gives nominal, unadjusted p-values 0.0338 (profiled) and 0.0637. These compare the two simulation methods, not the particle significance of the data. A nominal value below 0.05 is a warning about approximation error within this set of four comparisons for the ordering; it is not a definitive failure or a far-tail validation.

2021 10%: the distribution-shape (KS) diagnostic gives nominal, unadjusted p-values 0.5579 (profiled) and 0.4124. These compare the two simulation methods, not the particle significance of the data.

The direct-scan counts calibrate the declared score under the chosen background without requiring a jointly Gaussian field. The GP approximation has that additional assumption.

9 What this does and does not imply

A small conditional probability can reject the behavior predicted by one background construction without identifying the reason. In 2016, large null offsets make this distinction essential. An unusually high centered score and an ordinary raw maximum can coexist because the two rules ask different questions.

2016 full: the two 2 MeV subgrids give profiled global estimates $<1.50 \times 10^{-5}$ and $<1.50 \times 10^{-5}$, compared with $<1.50 \times 10^{-5}$ on the 1 MeV grid, all at the same fine-grid observed threshold. Zero-count comparisons are unresolved. A coarser-grid comparison cannot establish convergence to a continuous mass search.

2021 10%: the two 2 MeV subgrids give profiled global estimates 0.02 and 0.0202, compared with 0.0252 on the 1 MeV grid, all at the same fine-grid observed threshold. Zero-count comparisons are unresolved. A coarser-grid comparison cannot establish convergence to a continuous mass search.

Profiling and calibration answer different questions. Profiling accounts for the background freedom represented in the likelihood [2]. Calibration checks the statistic's sampling behavior. The global correction accounts for searching many masses. Passing one layer does not establish the other two, and none of these calculations by itself measures expected sensitivity or improves an upper limit.

The original calibration envelope is a separate target. The v4.9.13 local result takes an envelope over different background constructions. Here each complete scan uses one common spectrum. Neither new global ordering is a global calibration of that earlier envelope. The local GP generating models vary with mass and cannot simply be joined into one hypothetical experiment.

Dataset	Method	Raw peak [MeV]	GP global p	Direct exceedances
2016 full	Profiled	90	1	1000/1000
2016 full	Fixed	88	1	1000/1000
2021 10%	Profiled	78	0.7849	778/1000
2021 10%	Fixed	77	0.8845	877/1000

Table 7: Separate maximum-raw-root ordering, shown for comparison. A probability near one means that the selected background often produces a larger positive peak somewhere. It is not a goodness-of-fit test or evidence that the generating model is correct.

Next scientific steps. Qualify the 2016 background family using predeclared predictive controls, including its transition region. Compare fixed and reestimated kernel policies on the same spectra. Resolve the inherited numerical exception. Add direct scans where the tail of interest remains unresolved, and test a finer mass grid using a declared rule for intermediate kernel states. A joint result requires a shared-coupling likelihood and coherent joint background scenarios; multiplying or combining the individual probabilities here is not that calculation.

10 Reproducibility and references

All new numerical products are under

`study_results/v4p9p15_global_2016_2021_20260906/`.

The statistical procedure and spectrum seeds match the 2015 extension plan. A separately audited numerical accelerator makes the larger calculation practical. The byte-identical exact reference runner, all exact pilot scans and the partially completed exact 2016 validation remain available. Dataset-specific seeds separate experiments; pilot and validation seeds are also distinct. Source and count contracts bind every numerical ensemble. Every mass has a saved audit, including checks against the scalar fitter and the frozen observed result. No failed fits are removed. The independent HEP review is retained under `review/`.

Exact comparisons cover ten pilot scans at every mass, plus 1,000 scans at 81 of the 2016 masses. The 2021 validation ensemble uses the gated accelerator, with exact checks supplied by the pilot and response audits. Every mass has an exact baseline and bin-response stencil; six masses per dataset have complete exact response columns. 2016 full: 5 masses use the exact fallback; the largest paired-root difference is 3.7e-05; complete exact response columns at six masses differ by at most 8.3e-06 in relative vector norm, and their correlations by at most 5.5e-07. 2021 10%: 0 masses use the exact fallback; the largest paired-root difference is 1.2e-05; complete exact response columns at six masses differ by at most 1.3e-06 in relative vector norm, and their correlations by at most 4.6e-07. Numerical fits ran sequentially with one worker and one BLAS thread. The times below sum ensemble evaluations; shared exact accuracy gates and setup add to the total runtime. 2016 full: the accepted accelerated calculation, including accuracy checks, took 15.2 minutes. 2016 full: accelerated replay of the ten pilot scans took 4.2 s; 1,000 validation scans took 5.4 minutes; 721 Asimov scans took 3.8 minutes. 2021 10%: the accepted accelerated calculation, including accuracy checks, took 4.2 minutes. 2021 10%: accelerated replay of the ten pilot scans took 2.5 s; 1,000 validation scans took 2.2 minutes; 423 Asimov scans took 0.9 minutes. Timing is descriptive and depends on the host. The Gaussian field sampling and plots are a separate, inexpensive step.

Run these commands from the repository root. Replace `2016` with `2021` for the second dataset. Completed numerical phases resume only when their contracts agree.

```
nice -n 10 python3 -B \
  study_results/v4p9p15_global_2016_2021_20260906/run_global_accelerated.py \
  --dataset 2016
python3 -B \
  study_results/v4p9p15_global_2016_2021_20260906/analyze_global.py \
  --dataset 2016
```

The CSV curves, covariance matrices, complete scan vectors and GP maxima are retained for independent checks. `NEXT_STEPS.md` describes additional precision, finer grids and combined searches. `MANIFEST.csv` binds the final products; the earlier 2015 report and its manifest are checked without being rewritten.

References

[1] V. Ananiev and A. L. Read, “Gaussian Process-based calculation of look-elsewhere trials factor,” JINST **18** (2023) P05041. arXiv:2206.12328v3.

[2] G. Cowan, K. Cranmer, E. Gross and O. Vitells, “Asymptotic formulae for likelihood-based tests of new physics,” Eur. Phys. J. C **71** (2011) 1554. arXiv:1007.1727.